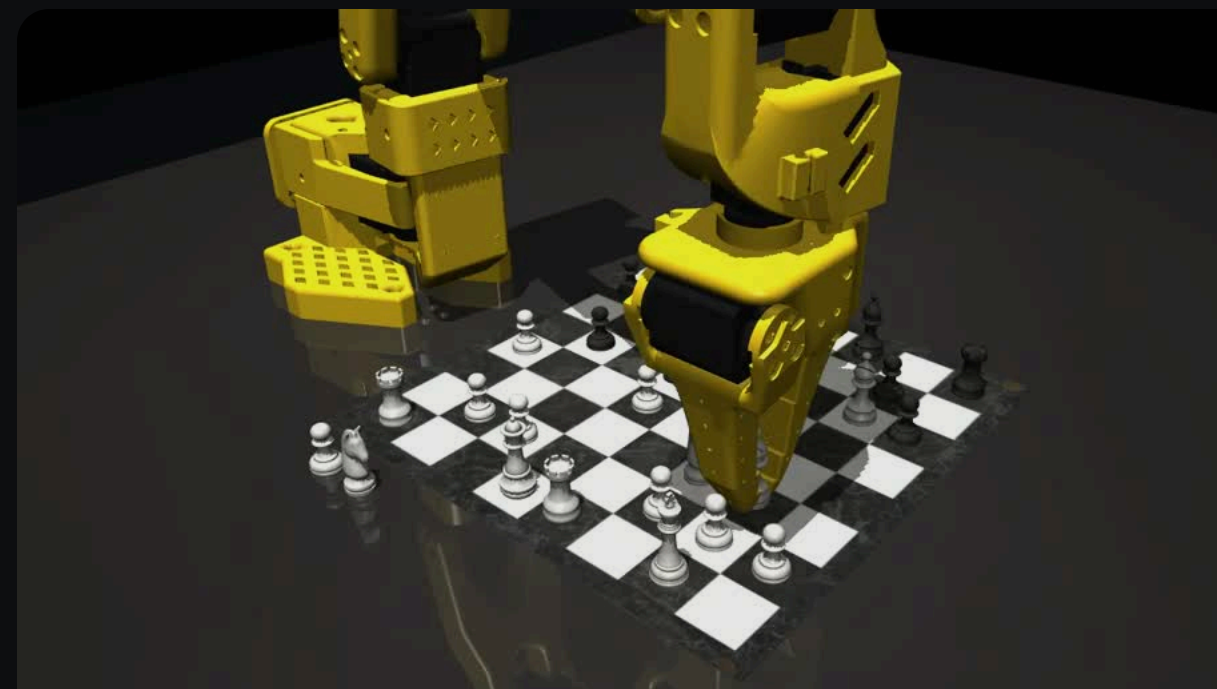


THE AGENT ARENA · FUTURE OF WORK: AI + ROBOTICS ON VULTR

Replay

Robot skills from one sentence,
learned from the open web.

Pranav Achar · S0-101 · MuJoCo · Vultr · replay-so101.vercel.app



THE PROBLEM

ChatGPT learned by reading the internet. Robots can't.

Every skill a robot has today, somebody recorded by hand: a person steering the arm, over and over, for one task. A second task means starting again.

The new robot models (VLAs) are starving for exactly this data. Meanwhile the internet is full of people showing how to do things, and full of the rules for doing them.

THE IDEA

Turn the open web into robot skills.

How people move lives in video. What the rules are lives in text.

Replay takes one sentence, finds both, and turns them into a skill for the SO-101 arm. Nothing reaches the robot until it has worked in physics.

HOW IT WORKS

One prompt, one robot, one sandbox.

1 · PLAN

An agent turns the sentence into a search plan.

2 · SCRAPE

Scrapling runs in a throwaway Vultr VM: Creative Commons video, web pages, licences checked.

3 · CHECK

A vision model checks each clip; python-chess checks every game move.

4 · TRACK

MediaPipe tracks the person's hand and body in every frame.

5 · RETARGET

The motion is mapped onto the SO-101's joints.

6 · MULTIPLY

Replayed on hundreds of new layouts in MuJoCo; only successes are kept (the MimicGen idea).

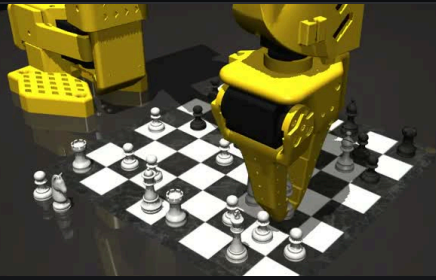
7 · TRAIN OR PLAN

A small policy learns by imitation (ACT-style chunking), or the moves are planned and rehearsed.

8 · TEST

Scored on layouts it has never seen, then downloadable.

Six skills, six different kinds of task.



Deep Blue vs. Kasparov, 1997, G6

37 moves, both sides · 47 pick-and-places ·
worst 4.9 mm



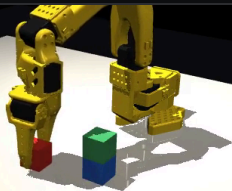
Dumbbell curl, copied 1:1

from Creative Commons video · elbow within
1.25°



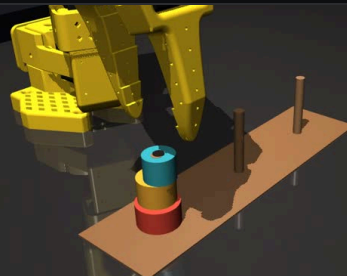
Block in the bowl

trained policy · 19/20 unseen layouts · 21 s
on CPU



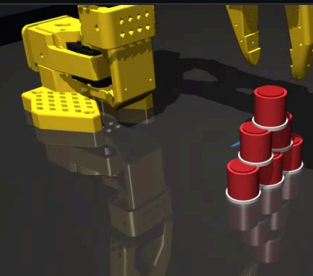
Block tower

trained policy · 18/20 unseen layouts



Tower of Hanoi

optimal 7 moves · each checked in physics ·
worst 6.9 mm



Cup pyramid

6 cups, 3-2-1 · worst 4.2 mm

Reading the open web, **safely**, on CPU.

- Every robot's scraping runs in its own Vultr VM that exists only for that run.
- Untrusted pages never touch the app: no secrets on the VM, strict output checks.
- The whole pipeline runs on CPU. No GPU anywhere.

NEXT

From simulation to the real arm.

- Hardware test on a real SO-101 (assembled, testing next week).
- Fine-tune SmolVLA, the open VLA built for this arm, on Replay's data.
- Try it: replay-so101.vercel.app · library: replay-library.vercel.app
- Code: github.com/PranavAchar01/agent-arena